

# EXPERIMENT-2

## STEP1: SOURCE CODE:-

The given source code contains standard input/output header , functions, macro/symbolic constant, and single – line comments.



```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~/task1
Session Actions Edit View Help
#include<stdio.h>           //preprocessor directive to include builtin header file
#define PI 3.14             //PPD to define a symbolic constant or macro
int main(void)
{
    int radius;             //variable definition
    float area;
    printf("Enter the radius"); //printf() is builtin function to display output string
    scanf("%d", &radius);    //scanf() BIF to get inputs
    area = PI*radius*radius; //compute
    printf ("Area = %f\n", area);
    return 0;
}
```

## STEP2: STAGES OF COMPILATION:-

- Preprocessing stage:  
The preprocessor prepares your C code before actual compilation. It mainly handles statements beginning with #.
- Compilation stage:  
Now the **compiler** processes the pre-processed C program. It analyses your C code and translates it into **assembly language**. The compiler also detects many programming errors, such as syntax errors:
- Assembling stage:  
The assembly language produced by the compiler and converts it into **machine code** and result is stored in an object file.
- Linking stage:  
The linker combines your object file with required library code to create the final executable program.

### STEP3: SAVING THE FILE AND EXECUTION:-

```
(kali㉿kali)-[~/task1]
$ gcc -save-temps second.c

(kali㉿kali)-[~/task1]
$ ls
a.out  a-second.i  a-second.o  a-second.s  second.c
```

For saving the file , we used: gcc -save-temps with the file.

a.out -The linker combines the object file with required startup code and libraries, resolves external references, and produces the runnable program.

### STEP4: FINAL RESULT

```
(kali㉿kali)-[~/task1]
$ ./a.out
Enter the radius12
Area = 452.160004
```